

Erica Mae C. Mallari BSIT 2-1 Midterm Activity #3

Code:

```

import java.util.Scanner;

//class name
public class ATMTransactionCounter Mallari {
    //Main: method (void)
    public static void main (String args[]) {
        Scanner input = new Scanner(System.in);

        float balance = 8000.00;
        boolean active = true;
        int transaction = 0;

        // Menu Displaying
        System.out.println("\n === ATM Menu ===");
        System.out.println("1. Check balance");
        System.out.println("2. Deposit");
        System.out.println("3. Withdraw");
        System.out.println("4. Exit\n");

        // While loop (menu)
        while (active) {
            System.out.print("\n Please select transaction: ");
            int option = input.nextInt();

            // switch case (menu)
            switch (option) {
                case 1: // Check balance
                    System.out.printf("Balance = %.2f\n", balance);
                    transaction++;
                    continue;

```


case 2: // Deposit amount

System.out.print("Deposit\n");

float deposit = input.nextFloat();

balance += deposit;

System.out.printf("New balance: %.2f\n", balance);

transaction++;

continue;

case 3: // Withdraw amount

boolean valid = true;

while (valid) {

System.out.print("Withdraw\n");

float withdraw = input.nextFloat();

System.out.print("Withdraw amount validation\n");

if (withdraw > balance) {

System.out.println("Error! Not enough balance.");

} else if (withdraw > 2000) {

System.out.println("Error! Maximum

withdrawal amount exceeded.");

} else {

balance -= withdraw;

valid = false;

}

}

System.out.printf("New balance: %.2f\n", balance);

System.out.print("Balance: %.2f\n", balance);

transaction++;

valid = true;

continue;

case 4: //Exit program

System.out.println("\n =====");
 System.out.println("Transaction Count: %d", transaction);

System.out.println("\n =====");
 System.out.println("\n Exiting program...");

active = false;

continue;

// Invalid answer message

System.out.println("Invalid input. Please select

from 1-4 only.");

continue;

// Close JAVA's scanner

input.close();

sample execution

Please enter transaction: 1

Balance = 8000.00

Please enter transaction: 2

Deposit = 500

New Balance = 8500.00

Please enter transaction: 3

Withdraw = 200

New Balance = 8300.00

Please enter transaction: 4

Transaction count: 3

=====

Exiting Program...

Explain:

1) How the while loop works

↳ The while loop executes the block of code inside it as long as the given parameter or expression is true. Before executing the block of code, it first checks if the expression is true. Unlike do-while which executes the code first before checking the expression.

2) How the switch case controls the menu

↳ The switch case controls the menu by directing the program to execute a certain block of code depending on the parameter.

For example, if the user chose "1", the code will direct the program to "case 1" and proceed with its code. Switch case is much easier to use and more neat than using if-else statements.

3) How if-else validates with drawal

↳ The program will check if the entered amount to be withdrawn exceeds the user's balance. Next, it will check if the amount exceeds P2000, the maximum withdrawal amount per transaction. If neither are true, it proceeds with 'else' - withdrawal. The program checks if it exceeds the balance first because the user may not have enough to withdraw but it's below 2000, so the program may flag it as valid. So, to avoid that from happening, I check if they have sufficient balance first.